Zero Trust Architecture for an Enterprise

Complete step-by-step build guide for Cisco Packet Tracer

Project Option 1: 802.1X port authentication, RADIUS AAA, VLAN micro-segmentation with ACLs, and Cisco ASA 5506-X security zones.

Part	Contents
1	Project brief: what we are building and why
2	Design: topology, addressing, devices, cabling, credentials
3	Build phases 1 to 13: click-by-click and command-by-command
4	Final tests, evidence checklist, troubleshooting, report tips

How to use this guide: follow the phases in order and do every checkpoint. Do not skip ahead. Each phase builds on the one before, and the checkpoints tell you the exact moment something went wrong so you never debug more than one thing at a time.

Part 1: Project Brief

1.1 What we are doing

We are designing and simulating an enterprise network that follows the **Zero Trust** security model, entirely inside Cisco Packet Tracer (CPT). Traditional networks use a "castle and moat" model: a strong firewall at the edge, and everything inside is trusted. Once an attacker gets in, they can move freely from machine to machine. Zero Trust replaces this with an **"assume breach"** approach: no device or user is trusted just because it is plugged into the internal network.

1.2 Objective

Prevent **lateral movement**. If an attacker compromises one endpoint, they must be contained to that segment and unable to reach servers, management systems or other departments.

1.3 The four security controls we implement

Control	What it does	CPT feature used
Access control	A device cannot use a switch port until the user proves who they are.	802.1X on the 2960 access switch
Identity management	A central server verifies credentials and tells the switch to open or keep closed the port.	AAA server (RADIUS) on a Server-PT
Micro-segmentation	The network is split into VLANs (Users, Finance, Servers, Management). Traffic between VLANs is denied unless explicitly allowed.	VLANs + SVIs + extended ACLs on the 3560 multilayer switch
Zone defense	A firewall separates Inside, DMZ and Outside. Lower-security zones cannot start connections to higher-security zones.	Cisco ASA 5506-X with security levels, NAT and ACLs

1.4 What we will prove (our demonstration)

- A device with no credentials, or the wrong password, gets no network access.
- An authenticated user can reach only what they need (for example, the internal web server on port 80), and nothing else.
- A Users PC cannot ping the Servers, Management or Finance VLANs, so a compromised PC cannot pivot.
- The DMZ web server can be reached from the Internet, but a compromised DMZ server cannot reach the inside network.
- The Internet cannot reach anything inside except the published DMZ web service.

1.5 Devices and tools needed

Qty	Device (as named in CPT)	Role
1	ASA 5506-X (Network Devices > Security)	Firewall: Inside / DMZ / Outside zones
1	3560-24PS multilayer switch	Core: VLAN gateways, ACL enforcement
1	2960-24TT switch	Access switch: 802.1X ports
1	2911 router (or 1941)	ISP / Internet simulation
3	Server-PT	AAA (RADIUS), internal web, DMZ web

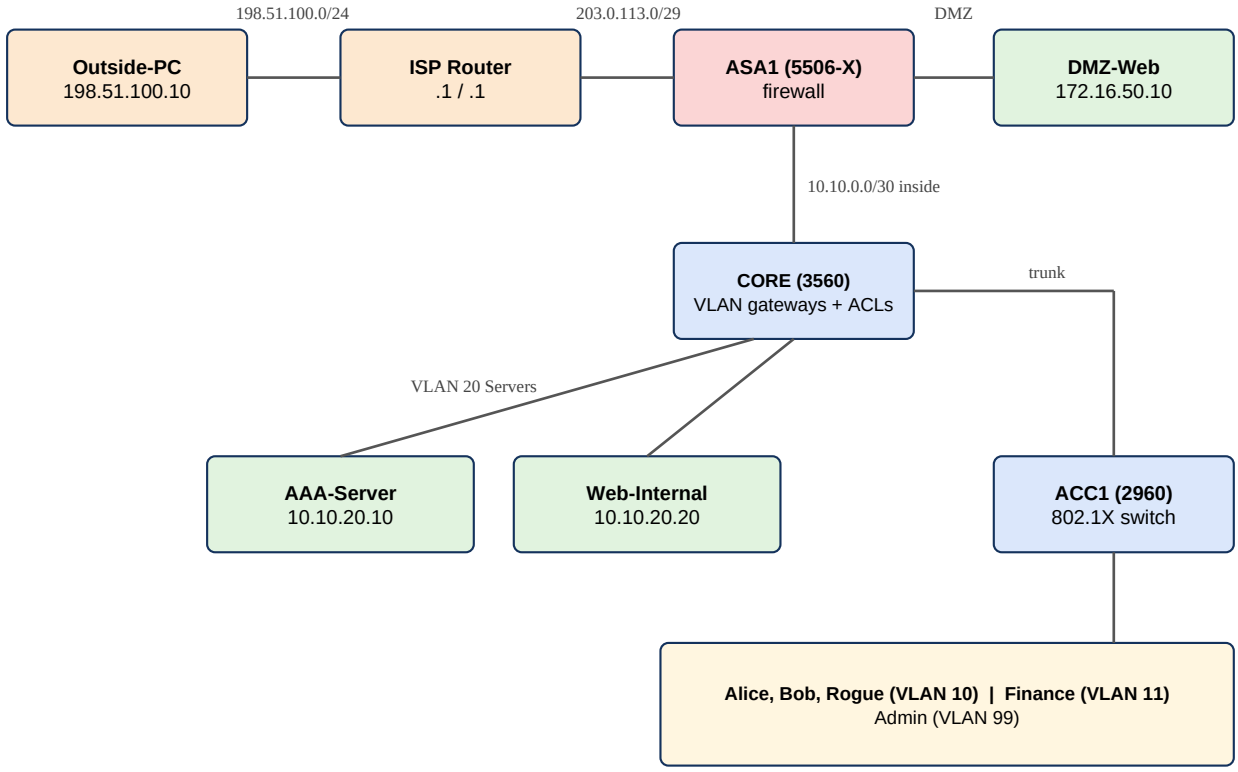
Qty	Device (as named in CPT)	Role
6	PC-PT	Alice, Bob, Rogue, Finance, Admin, Outside
-	Copper straight-through and cross-over cables	Connectivity

1.6 Honest limitations (mention these in your report)

- Hosts in the same VLAN can still reach each other, because CPT has no private VLANs. We reduce this by giving departments their own VLANs.
- CPT 802.1X is basic: MD5 username/password only. No certificates, no dynamic VLAN assignment from RADIUS, no posture checks.
- The CPT ASA is a reduced feature set: no ASDM and limited inspection features.
- Command syntax can differ slightly by CPT version. If a command is rejected, type the beginning followed by ? and use the closest option.

Part 2: Design

2.1 Topology



2.2 Addressing plan

Zone / VLAN	Subnet	Gateway	Hosts
VLAN 10 Users	10.10.10.0/24	10.10.10.1	Alice .11, Bob .12, Rogue .66
VLAN 11 Finance	10.10.11.0/24	10.10.11.1	Fin-PC .11
VLAN 20 Servers	10.10.20.0/24	10.10.20.1	AAA .10, Web-Internal .20
VLAN 99 Management	10.10.99.0/24	10.10.99.1	ACC1 .11, Admin-PC .50
Core to ASA link	10.10.0.0/30	-	CORE .2, ASA inside .1
DMZ	172.16.50.0/24	172.16.50.1 (ASA)	DMZ-Web .10
Outside link	203.0.113.0/29	-	ISP .1, ASA outside .2, DMZ public (NAT) .3
Internet side	198.51.100.0/24	198.51.100.1 (ISP)	Outside-PC .10

2.3 Credentials and secrets

Item	Value
RADIUS shared secret	Secret123
alice (Alice-PC)	Pass123
bob (Bob-PC)	Pass456
carol (Fin-PC)	Pass789

Item	Value
Rogue-PC	No credentials (this is our attacker device)

2.4 Cable and port map

Use **Copper Straight-Through** unless the table says **Cross-Over**. (Rule of thumb: like devices such as switch to switch, router to PC, or ASA to router use cross-over.)

From (device : port)	To (device : port)	Cable
Outside-PC : FastEthernet0	ISP : GigabitEthernet0/1	Cross-Over
ISP : GigabitEthernet0/0	ASA1 : GigabitEthernet1/3	Cross-Over
ASA1 : GigabitEthernet1/2	DMZ-Web : FastEthernet0	Cross-Over
ASA1 : GigabitEthernet1/1	CORE : GigabitEthernet0/1	Straight
CORE : FastEthernet0/1	AAA-Server : FastEthernet0	Straight
CORE : FastEthernet0/2	Web-Internal : FastEthernet0	Straight
CORE : GigabitEthernet0/2	ACC1 : GigabitEthernet0/1	Cross-Over
ACC1 : FastEthernet0/1	Alice-PC : FastEthernet0	Straight
ACC1 : FastEthernet0/2	Bob-PC : FastEthernet0	Straight
ACC1 : FastEthernet0/3	Rogue-PC : FastEthernet0	Straight
ACC1 : FastEthernet0/6	Fin-PC : FastEthernet0	Straight
ACC1 : FastEthernet0/11	Admin-PC : FastEthernet0	Straight

> **NOTE:** If your ASA has different port names, use the same positions (first port = inside, second = DMZ, third = outside) and change the interface names in Phase 7 to match. Hover over a port while cabling to read its exact name.

Part 3: Build Phases

Estimated total time: 2 to 3 hours the first time. Save your file often (File > Save, shortcut Ctrl+S). Name it `ZeroTrust.pkt`.

Phase 1: Prepare and Test 802.1X Feasibility

1.1 Open Cisco Packet Tracer. Check the version at **Help > About**. Version 8.x or newer is recommended.

1.2 Save a new file straight away: **File > Save As**, name it `ZeroTrust.pkt`.

1.3 **Quick 802.1X test (15 minutes).** In the bottom-left device panel click **Network Devices** (icon of a router and switch), then **Switches**, and drag a **2960** onto the canvas. Click **End Devices**, drag one **PC** and one **Server** (Server-PT) onto the canvas.

1.4 Connect them: click the **Connections** icon (lightning bolt at the bottom of the device panel), choose **Copper Straight-Through**, click the PC > FastEthernet0, then click the switch > FastEthernet0/1. Repeat for the server on FastEthernet0/2.

1.5 Give the server IP `10.0.0.10/24` and the PC `10.0.0.11/24` (see Phase 3 for how to set IPs). On the server open the **Services** tab > **AAA**, set Service **On**, add client `SW`, IP `10.0.0.2`, secret `Secret123`, ServerType **Radius**, click **Add**. Add user `test` / `test`.

1.6 Open the switch **CLI** tab, press Enter, and type:

```
enable
configure terminal
interface vlan 1
 ip address 10.0.0.2 255.255.255.0
 no shutdown
aaa new-model
radius-server host 10.0.0.10 key Secret123
aaa authentication dot1x default group radius
dot1x system-auth-control
interface fastethernet0/1
 switchport mode access
 dot1x port-control auto
end
```

1.7 On the PC: **Config** tab > **FastEthernet0**. Find the **Authentication** box, tick 802.1X, protocol MD5, username `test`, password `test`. Without the credentials the PC must not be able to ping the server; with them it should.

> **CHECKPOINT:** If the PC has no Authentication box, or the port never authorizes, STOP and update to a newer CPT version before continuing. If it works, delete these three test devices (select, press Delete) and begin Phase 2 on a clean canvas.

Phase 2: Place the Devices

All devices are placed from the device panel in the bottom-left corner of CPT. Click a category icon in the top row, a sub-category in the second row, then the device in the list. Drag it onto the canvas.

2.1 Click **Network Devices** > **Security** > drag the **ASA 5506-X** to the center of the canvas.

2.2 Click **Network Devices** > **Switches** > drag a **3560-24PS** below the ASA. This is the CORE switch.

2.3 Still in **Switches**, drag a **2960-24TT** to the lower right. This is ACC1.

2.4 Click **Network Devices** > **Routers** > drag a **2911** to the left of the ASA. This is the ISP router.

2.5 Click **End Devices** > drag **three Server-PT** devices: two near the CORE switch (AAA-Server, Web-Internal) and one to the right of the ASA (DMZ-Web).

2.6 In **End Devices** drag **six PC-PT**: Alice-PC, Bob-PC, Rogue-PC, Fin-PC, Admin-PC near ACC1 and Outside-PC at the far left near the ISP router.

2.7 **Rename each device.** Click a device, open the **Config** tab, and type the name in **Display Name** (for example `Alice-PC`). Use the names from the cable table in Section 2.4: ISP, ASA1, CORE, ACC1, AAA-Server, Web-Internal, DMZ-Web, Alice-PC, Bob-PC, Rogue-PC, Fin-PC, Admin-PC, Outside-PC. Hostnames on the switches, router and ASA are set again in the CLI later.

> **CHECKPOINT:** You should have 1 ASA, 2 switches, 1 router, 3 servers and 6 PCs on the canvas, all named.

Phase 3: Connect the Cables

3.1 Click the **Connections** icon (yellow lightning bolt) at the bottom of the device panel.

3.2 For each row of the cable table in Section 2.4: choose the cable type (**Copper Straight-Through** is the solid black line, **Copper Cross-Over** is the dashed line), click the first device and choose the listed port from the pop-up, then click the second device and choose its port.

3.3 Work top to bottom through the table so you do not miss any. After each cable, a colored dot appears at each end: **red** means the link is down, **green** means it is up, **amber** means the switch port is still starting.

3.4 Click the **Fast Forward Time** button (two right arrows, bottom-left, in the time bar) a few times to speed up the port start-up. Amber dots turn green in about 30 simulated seconds.

> **NOTE:** Some links, such as ASA to ISP, stay red until the interfaces are enabled in later phases. That is normal. Switch-to-PC links should already be green or amber.

> **CHECKPOINT:** Every switch-to-PC and switch-to-server link is green or amber. Save the file.

Phase 4: Set IP Addresses on All End Devices

For **every PC and server**: click the device, open the **Desktop** tab, click **IP Configuration**, choose **Static**, and type the values below. Close the window when done.

Device	IP address	Subnet mask	Default gateway
Alice-PC	10.10.10.11	255.255.255.0	10.10.10.1
Bob-PC	10.10.10.12	255.255.255.0	10.10.10.1
Rogue-PC	10.10.10.66	255.255.255.0	10.10.10.1
Fin-PC	10.10.11.11	255.255.255.0	10.10.11.1
Admin-PC	10.10.99.50	255.255.255.0	10.10.99.1
AAA-Server	10.10.20.10	255.255.255.0	10.10.20.1
Web-Internal	10.10.20.20	255.255.255.0	10.10.20.1
DMZ-Web	172.16.50.10	255.255.255.0	172.16.50.1
Outside-PC	198.51.100.10	255.255.255.0	198.51.100.1

4.1 Click **Web-Internal** > **Services** tab > **HTTP**. Make sure HTTP is **On**. Click **index.html** > **(edit)** and change the text to `INTERNAL WEB SERVER`, then **Save**.

4.2 Do the same on **DMZ-Web** and change the text to `DMZ WEB SERVER`. This lets you tell the two servers apart in later browser tests.

> **CHECKPOINT:** Each device's IP Configuration window shows the right address. Do not worry that nothing can ping yet.

Phase 5: Configure the ISP Router

5.1 Click **ISP** > **CLI** tab. If asked `Would you like to enter the initial configuration dialog?`, type `no` and press Enter, then press Enter again for the prompt.

5.2 Type the following exactly:

```
enable
configure terminal
hostname ISP
interface gigabitethernet0/0
 description To ASA outside
 ip address 203.0.113.1 255.255.255.248
 no shutdown
interface gigabitethernet0/1
 description To Outside-PC
 ip address 198.51.100.1 255.255.255.0
 no shutdown
end
write memory
```

CHECKPOINT: `show ip interface brief` shows both interfaces up/up (Gi0/0 may show down until the ASA is configured). From Outside-PC: Desktop > Command Prompt > `ping 198.51.100.1` succeeds.

Phase 6: Configure the CORE Switch (3560)

This switch is the gateway for every internal VLAN. The ACLs are added later in Phase 12, so connectivity can be tested first.

6.1 Click **CORE** > **CLI** tab. Press Enter to get the prompt.

6.2 Create the VLANs and the gateway interfaces (SVIs):

```
enable
configure terminal
hostname CORE
ip routing
vlan 10
 name USERS
vlan 11
 name FINANCE
vlan 20
 name SERVERS
vlan 99
 name MGMT
exit
interface vlan 10
 ip address 10.10.10.1 255.255.255.0
 no shutdown
interface vlan 11
 ip address 10.10.11.1 255.255.255.0
 no shutdown
interface vlan 20
 ip address 10.10.20.1 255.255.255.0
 no shutdown
interface vlan 99
 ip address 10.10.99.1 255.255.255.0
 no shutdown
```

6.3 Configure the routed link to the ASA and the default route:

```
interface gigabitethernet0/1
 description To ASA inside
 no switchport
 ip address 10.10.0.2 255.255.255.252
 no shutdown
ip route 0.0.0.0 0.0.0.0 10.10.0.1
```

6.4 Configure the trunk to ACC1 and the two server ports:

```
interface gigabitethernet0/2
 description Trunk to ACC1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 no shutdown
interface fastethernet0/1
 description AAA-Server
 switchport mode access
 switchport access vlan 20
interface fastethernet0/2
 description Web-Internal
 switchport mode access
 switchport access vlan 20
end
write memory
```

> **CHECKPOINT:** `show vlan brief` lists VLANs 10, 11, 20, 99, with Fa0/1 and Fa0/2 in VLAN 20. `show ip interface brief` shows Vlan20 up/up. (Vlan10, 11 and 99 turn up once ACC1 is configured in Phase 7. This is normal.)

Phase 7: Configure the Access Switch (ACC1)

In this phase we only configure VLANs, the trunk and port assignments. 802.1X is added later in Phase 10, so we can test basic connectivity first.

7.1 Click **ACC1** > **CLI** tab and type:

```
enable
configure terminal
hostname ACC1
vlan 10
 name USERS
vlan 11
 name FINANCE
vlan 20
 name SERVERS
vlan 99
 name MGMT
exit
interface vlan 99
 ip address 10.10.99.11 255.255.255.0
 no shutdown
ip default-gateway 10.10.99.1
interface gigabitethernet0/1
 description Trunk to CORE
 switchport mode trunk
```

7.2 Assign the access ports to their VLANs:

```
interface range fastethernet0/1-5
 switchport mode access
 switchport access vlan 10
interface range fastethernet0/6-10
 switchport mode access
 switchport access vlan 11
interface fastethernet0/11
 switchport mode access
 switchport access vlan 99
interface range fastethernet0/12-24
 shutdown
end
write memory
```

> **CHECKPOINT:** `show interfaces trunk` shows Gi0/1 trunking. `show vlan brief` shows Fa0/1-5 in VLAN 10, Fa0/6-10 in VLAN 11, Fa0/11 in VLAN 99. From ACC1: `ping 10.10.99.1` succeeds (the first reply may time out; try twice).

Phase 8: Configure the ASA 5506-X Firewall

The ASA is the zone-defense device. Each interface gets a name and a security level: inside 100 (most trusted), dmz 50, outside 0 (least trusted). By default the ASA lets higher levels start traffic toward lower levels, and blocks the reverse.

8.1 Click **ASA1** > **CLI** tab. At the `ciscoasa>` prompt type `enable`. When asked for a password just press Enter (the password is blank). Then type `configure terminal`.

8.2 Set the hostname and the three interfaces:

```
hostname ASA1
interface gigabitethernet1/1
 nameif inside
 security-level 100
 ip address 10.10.0.1 255.255.255.252
 no shutdown
interface gigabitethernet1/2
 nameif dmz
 security-level 50
 ip address 172.16.50.1 255.255.255.0
 no shutdown
interface gigabitethernet1/3
 nameif outside
 security-level 0
 ip address 203.0.113.2 255.255.255.248
 no shutdown
exit
```

8.3 Add the routes: default route to the ISP and a route to the internal networks via the CORE switch:

```
route outside 0.0.0.0 0.0.0.0 203.0.113.1
route inside 10.10.0.0 255.255.0.0 10.10.0.2
```

8.4 Add NAT. Inside users share the outside interface address (PAT), and the DMZ web server is published as 203.0.113.3:

```
object network INSIDE_NET
 subnet 10.10.0.0 255.255.0.0
 nat (inside,outside) dynamic interface
object network DMZ_WEB
 host 172.16.50.10
 nat (dmz,outside) static 203.0.113.3
exit
```

8.5 Add the firewall rules. The outside may reach only the DMZ web server on port 80. The DMZ may not start anything toward the inside:

```
access-list OUTSIDE_IN extended permit tcp any host 172.16.50.10 eq 80
access-group OUTSIDE_IN in interface outside
access-list DMZ_IN extended deny ip 172.16.50.0 255.255.255.0 10.10.0.0 255.255.0.0
access-list DMZ_IN extended permit ip any any
access-group DMZ_IN in interface dmz
```

8.6 Turn on ICMP inspection so ping replies from lower zones are allowed back (otherwise working pings look broken):

```
policy-map global_policy
 class inspection_default
  inspect icmp
exit
service-policy global_policy global
end
write memory
```

NOTE: If CPT rejects the `nat` lines inside the object (older versions), try `?` to see accepted syntax. If NAT cannot be made to work, remove the two NAT objects and instead add static routes on the ISP router for 10.10.0.0/16 and 172.16.50.0/24 via 203.0.113.2. Outside access to the DMZ then uses 172.16.50.10 directly and the ACL stays the same.

CHECKPOINT: `show interface ip brief` shows three interfaces up/up with the right addresses. `show route` lists the default route and 10.10.0.0/16. From ASA1: `ping 203.0.113.1`, `ping 10.10.0.2` and `ping 172.16.50.10` all succeed.

Phase 9: Baseline Connectivity Test (before security)

Security is not enabled yet, so almost everything should work. This proves the plumbing is correct, so any failure later is caused by security rules and not by a cabling or addressing error.

9.1 Alice-PC > **Desktop > Command Prompt**. Run these pings one at a time and note the results:

```
ping 10.10.10.1        (her gateway)          expect: success
ping 10.10.20.20       (internal web)         expect: success
ping 10.10.11.11       (Fin-PC)               expect: success
ping 10.10.99.50       (Admin-PC)             expect: success
ping 172.16.50.10      (DMZ-Web)              expect: success
ping 198.51.100.10     (Outside-PC)           expect: success
```

9.2 Outside-PC > Desktop > **Web Browser** > type `http://203.0.113.3` and press Go. You should see `DMZ WEB SERVER`. (If you used the no-NAT fallback, browse to `http://172.16.50.10`.)

> **CHECKPOINT:** All of the above work. If any fail, fix it now using the Troubleshooting table at the end. Pings may need a second try because the first one triggers ARP. Save the file.

Phase 10: Configure the AAA (RADIUS) Server

10.1 Click **AAA-Server** > **Services** tab > **AAA** in the left list.

10.2 Set **Service** to **On**.

10.3 Under **Network Configuration** fill in: Client Name `ACC1`, Client IP `10.10.99.11`, Secret `Secret123`, ServerType **Radius**. Click **Add**. The client appears in the table below.

10.4 Under **User Setup** add three users, clicking **Add** after each: `alice` / `Pass123`, `bob` / `Pass456`, `carol` / `Pass789`.

> **WARNING:** The Client IP must be exactly the address ACC1 uses to talk to the server: its VLAN 99 interface, 10.10.99.11. A mismatch is the number one reason 802.1X fails.

> **CHECKPOINT:** The client table shows one row (ACC1) and the user table shows three users.

Phase 11: Enable 802.1X on the Access Switch

11.1 Click **ACC1** > **CLI** tab and type:

```
enable
configure terminal
aaa new-model
radius-server host 10.10.20.10 key Secret123
aaa authentication dot1x default group radius
dot1x system-auth-control
interface range fastethernet0/1-10
 dot1x port-control auto
end
write memory
```

> **NOTE:** If `dot1x port-control auto` is rejected, use `authentication port-control auto` instead.

11.2 Port Fa0/11 (Admin-PC) is intentionally left without 802.1X in this design, so the administrator always has a way to reach the network if RADIUS fails. State this in your report as a deliberate break-glass choice.

11.3 **Configure the PC credentials.** Click **Alice-PC** > **Config** tab > **FastEthernet0**. Scroll to the **Authentication** section, tick **802.1X**, set Protocol to **MD5**, username `alice`, password `Pass123`.

11.4 Repeat for **Bob-PC** (username `bob`, password `Pass456`) and **Fin-PC** (username `carol`, password `Pass789`).

11.5 **Rogue-PC**: leave Authentication empty. This is the unauthorized device.

11.6 Wait for the port to re-authenticate: fast-forward time a few times, or on the PC un-tick then re-tick 802.1X to trigger a new attempt.

> **CHECKPOINT:** On ACC1 run `show dot1x all` (or `show dot1x`). Fa0/1, Fa0/2 and Fa0/6 show Authorized. Fa0/3 (Rogue-PC) shows Unauthorized. Ping tests: Alice-PC to 10.10.10.1 succeeds, Rogue-PC to 10.10.10.1 fails. Also test a wrong password: change Bob's password on the PC and confirm the port goes Unauthorized, then restore it.

Phase 12: Apply Micro-segmentation ACLs on the CORE Switch

Each VLAN gets an inbound ACL on its gateway interface (SVI). The design principle: users may reach only the internal web server on port 80 and the outside world; servers and management never initiate connections toward users; and the Users and Finance VLANs cannot reach each other.

12.1 Click **CORE** > **CLI** tab. Create the ACL for the Users VLAN:

```
enable
configure terminal
ip access-list extended USERS_IN
 permit tcp 10.10.10.0 0.0.0.255 host 10.10.20.20 eq 80
 deny   ip 10.10.10.0 0.0.0.255 10.10.0.0 0.0.255.255
 permit ip 10.10.10.0 0.0.0.255 any
exit
```

12.2 Create the ACL for the Finance VLAN (same pattern):

```
ip access-list extended FIN_IN
 permit tcp 10.10.11.0 0.0.0.255 host 10.10.20.20 eq 80
 deny   ip 10.10.11.0 0.0.0.255 10.10.0.0 0.0.255.255
 permit ip 10.10.11.0 0.0.0.255 any
exit
```

12.3 Create the ACL for the Servers VLAN. Servers may only send replies (established TCP), ping replies, and the RADIUS answers to the management VLAN:

```
ip access-list extended SERVERS_IN
 permit udp host 10.10.20.10 10.10.99.0 0.0.0.255
 permit tcp 10.10.20.0 0.0.0.255 any established
 permit icmp 10.10.20.0 0.0.0.255 any echo-reply
 deny   ip any any
exit
```

12.4 Create the ACL for the Management VLAN. The switch may talk RADIUS to the AAA server, and only the Admin PC has full access:

```
ip access-list extended MGMT_IN
 permit udp 10.10.99.0 0.0.0.255 host 10.10.20.10 eq 1812
 permit udp 10.10.99.0 0.0.0.255 host 10.10.20.10 eq 1645
 permit ip host 10.10.99.50 any
 deny   ip any any
exit
```

12.5 Apply each ACL inbound on its VLAN interface:

```
interface vlan 10
 ip access-group USERS_IN in
interface vlan 11
 ip access-group FIN_IN in
interface vlan 20
 ip access-group SERVERS_IN in
interface vlan 99
 ip access-group MGMT_IN in
end
write memory
```

WARNING: Do not remove the two RADIUS permit lines in MGMT_IN or the SERVERS_IN UDP line. Without them ACC1 cannot reach the AAA server and every 802.1X login fails.

CHECKPOINT: `show ip access-lists` lists all four ACLs. `show run | include access-group` shows four applied lines. Re-run the 802.1X check from Phase 10: authorized ports must stay authorized.

Phase 13: Run the Zero Trust Tests

Run every test below, record the result, and take a screenshot of each. Use the Command Prompt (ping) and Web Browser on each PC.

#	Test (from, action)	Expected result	Why
1	Rogue-PC pings 10.10.10.1	FAIL	802.1X: no credentials, port unauthorized
2	Change Bob's PC password to a wrong one, re-authenticate, ping 10.10.10.1	FAIL	RADIUS rejects, port stays closed
3	Alice-PC pings 10.10.10.1	OK	Valid user, port authorized
4	Alice-PC browses http://10.10.20.20	OK (INTERNAL WEB SERVER)	Port 80 explicitly permitted
5	Alice-PC pings 10.10.20.20	FAIL	ICMP to Servers VLAN denied
6	Alice-PC pings 10.10.20.10 (AAA)	FAIL	Users cannot touch the AAA server
7	Alice-PC pings 10.10.99.11 (ACC1) and 10.10.99.50	FAIL	Users cannot reach Management
8	Alice-PC pings 10.10.11.11 (Fin-PC)	FAIL	Lateral movement blocked between departments
9	Fin-PC pings 10.10.10.11 (Alice-PC)	FAIL	Blocked in the opposite direction too
10	Alice-PC pings 172.16.50.10 (DMZ)	OK	Inside to DMZ allowed by ASA
11	DMZ-Web pings 10.10.10.11	FAIL	ASA blocks DMZ to Inside
12	Outside-PC browses http://203.0.113.3	OK (DMZ WEB SERVER)	Published service permitted
13	Outside-PC pings 203.0.113.3	FAIL	Only TCP 80 permitted on outside ACL
14	Outside-PC pings 10.10.10.11	FAIL	No route/NAT to inside
15	Admin-PC pings 10.10.20.10, 10.10.20.20 and 10.10.99.11	OK	Admin has the one allowed management path
16	Alice-PC pings 198.51.100.10 (Internet)	OK	Inside to outside allowed by PAT

Show commands to capture as evidence

```
CORE# show ip access-lists          (hit counters prove traffic was denied)
CORE# show vlan brief
ACC1# show dot1x all                (Authorized / Unauthorized per port)
ASA1# show access-list              (hit counts on OUTSIDE_IN and DMZ_IN)
ASA1# show nat
ASA1# show xlate
```

NOTE: For test 11, click DMZ-Web > Desktop > Command Prompt and run `ping 10.10.10.11`. After it fails, run `show access-list` on the ASA and point to the DMZ_IN deny hit count.

Phase 14: Save Everything

14.1 On each CLI device (ISP, CORE, ACC1, ASA1) run `write memory` (or `copy running-config startup-config` on the switches and router).

14.2 Save the Packet Tracer file with **File > Save**.

14.3 Optional: use the **Add Simple PDU** tool (closed envelope icon on the right toolbar) for visual pass/fail scenarios. Click the envelope, then the source device, then the destination. The Event List at the bottom right shows Successful or Failed with a red X.

Part 4: Troubleshooting, Evidence and Report Tips

4.1 Troubleshooting

Symptom	Likely cause	Fix
Link dot stays red	Wrong cable type or wrong port	Delete the cable and use the cable-table type. Swap to cross-over for switch-to-switch and router-to-ASA.
First ping times out, later pings work	ARP learning	Normal. Ping a second time.
Ping to gateway fails	VLAN missing, port in wrong VLAN, SVI down	`show vlan brief` on both switches. Check `show ip interface brief` on CORE.
SVI Vlan99 stays down/down	ACC1 not yet configured or trunk not up	Configure ACC1 and verify `show interfaces trunk`.
Cannot reach outside or DMZ	Missing default route on CORE, or ASA route	`show ip route` on CORE for 0.0.0.0/0 via 10.10.0.1. `show route` on ASA.
802.1X port stays Unauthorized for valid users	Client IP or secret mismatch on AAA server	Client IP must be 10.10.99.11. Secret must be identical on both sides. Re-add the client.
802.1X worked, then broke after the ACLs	RADIUS permit lines missing	Check MGMT_IN lines for 1812/1645 and the UDP permit in SERVERS_IN.
PC has no Authentication box	Older CPT version	Update to CPT 8.x.
Outside cannot reach DMZ web	NAT or OUTSIDE_IN ACL problem	`show nat`, `show xlate`, `show access-list`. Or use the no-NAT fallback from Phase 7.
Replies from DMZ or outside are blocked	ICMP inspection not enabled	Re-enter the policy-map block and `service-policy global_policy global`.
Command rejected	Syntax differs in your CPT version	Type the first part of the command and `?` to see valid options.

4.2 Evidence checklist (screenshots for your report)

- Final topology with all devices labeled and all links green.
- AAA server screen showing the client and the three users.
- `show dot1x all` output with Fa0/1, Fa0/2, Fa0/6 Authorized and Fa0/3 Unauthorized.
- Rogue-PC failed ping, and the wrong-password failure.
- Alice-PC web page success (port 80) and failed ping to the same server.
- Failed pings from Users to Servers, Management and Finance VLANs.
- `show ip access-lists` with non-zero deny counters.
- ASA `show access-list` and `show nat` output.
- Outside-PC browsing to the DMZ web server, and the failed ping to inside hosts.

4.3 Report tips

- Explain the castle-and-moat model first, then show how each control (802.1X, AAA, VLAN ACLs, ASA zones) removes a form of implicit trust.

- Use a table of attack scenarios: rogue device plugs in, compromised user PC scans the network, compromised DMZ server tries to pivot, Internet scans the inside. For each, state which control stops it and which test proves it.
- Be upfront about limitations (Section 1.6). Markers reward honest analysis, for example noting that same-VLAN hosts can still talk to each other and that production networks would add private VLANs, dynamic VLAN assignment and certificates.
- Optional extension if time allows: add a second Users PC pair in VLAN 10 and discuss how `switchport protected` (if your CPT version accepts it) or per-user VLANs would address same-VLAN movement.